Equations used in the model

The kinetic energy of a body is E = 1/2 m v^2, where m is the mass and v the velocity.

The area of a circle is A = π r^2.

The roots of a x^2 + b x + c = 0 are x = (-b ± √(b^2 - 4ac)) / 2a.

The sum of the first n integers is n(n+1)/2.